LAB REPORT: AGILE DEVELOPMENT & JIRA PROJECT MANAGEMENT

Project Title: EV Charging Station Management System

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1. Executive Summary

This report documents the application of Agile software development practices and Jira project management tools to design and execute a system for managing EV Charging Stations. The project breaks down core functional requirements into structured Epics and User Stories, applies story point estimations, and executes simulated sprints to track team velocity and burndown performance.

2. Epics & User Story Specifications

Epic 1: Connector Matching & Reservation

Description: Enables EV drivers to search and filter charging stations based on hardware compatibility with their vehicle (CCS, CHAdeMO, Type 2) and manage 30-minute reservation windows to secure charging slots in advance.

- **Story 1.1: Filter Chargers by Connector Type**
    - *Description*: As an EV Driver, I want to filter available charging stations by my vehicle's specific connector type (CCS, CHAdeMO, Type 2), so that I only navigate to stations that can physically charge my car.
- **Story 1.2: Reserve a 30-Minute Time Slot**
    - *Description*: As an EV Driver, I want to reserve a specific charger socket for a 30-minute window in advance, so that I am guaranteed a spot when I arrive.
- **Story 1.3: Extend Active Reservation**
    - *Description*: As an EV Driver, I want to request a 15-minute reservation extension from the app, so that I do not lose my booking if I run into light traffic.

Epic 2: Charging Station Discovery & Navigation

Description: Provides real-time mapping, dynamic pricing based on power demand, and turn-by-turn navigation guidance to help drivers discover, choose, and reach available charging stations efficiently.

- **Story 2.1: View Live Station Availability & Dynamic Rates**
    - *Description*: As an EV Driver, I want to see real-time charger availability and current peak/off-peak pricing on a map, so that I can choose the most cost-effective and convenient station.
- **Story 2.2: GPS Turn-by-Turn Navigation**
    - *Description*: As an EV Driver, I want to launch turn-by-turn navigation directly to my reserved station from the app, so that I can reach the charger without switching apps manually.

- **Story 2.3: Save Favorite Stations**
    - *Description*: As an EV Driver, I want to save frequently used charging stations to my favorites list, so that I can quickly check their availability and book them in the future.

Epic 3: Payment Processing & Booking Confirmation

Description: Manages secure digital transactions to confirm reservations, automated session billing, receipt generation, and rule enforcement for idle or unfulfilled bookings.

- **Story 3.1: Digital Payment Authorization**
    - *Description*: As an EV Driver, I want to pay for my reservation using digital payment methods (credit card, mobile wallet), so that my slot is instantly confirmed.
- **Story 3.2: Automated No-Show Release and Penalty**
    - *Description*: As a System Administrator, I want the system to automatically release a socket and issue a no-show fee if the car is not plugged in within 10 minutes of arrival time, so that idle chargers are freed up for other users.
- **Story 3.3: Digital Charging Receipt**
    - *Description*: As an EV Driver, I want to receive a digital receipt via email/app after a charging session finishes, so that I can track my spending and total energy delivered.

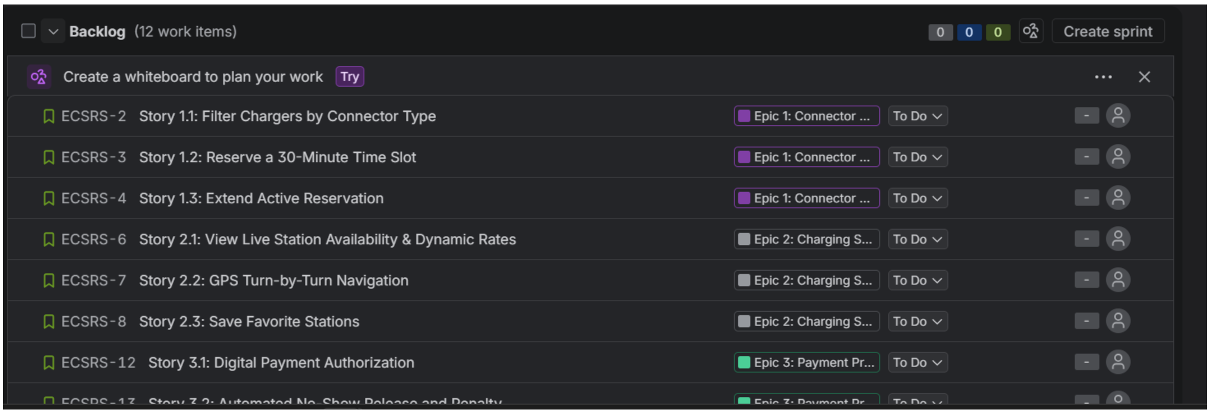
Epic 4: Station Operator Overrides & Maintenance

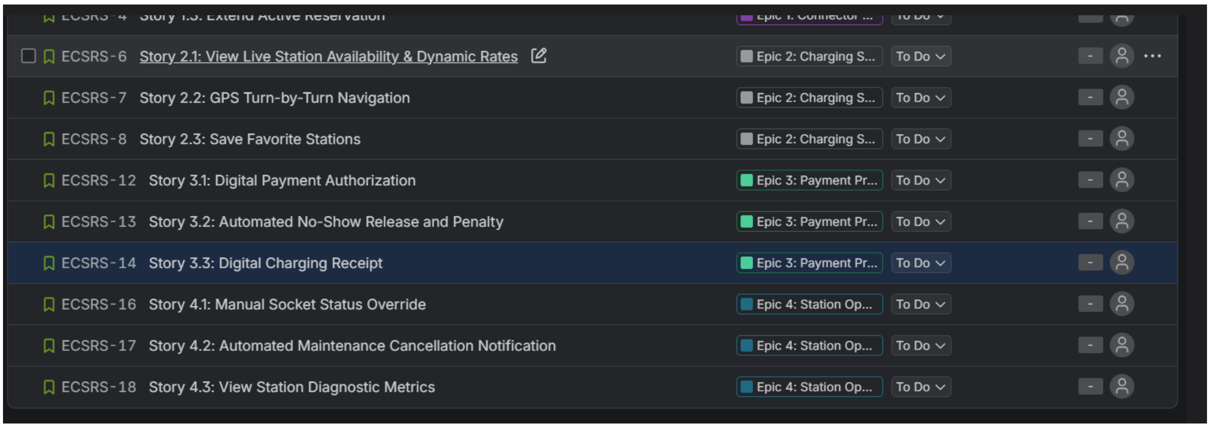
Description: Equips station operators and admins with override controls to modify hardware status, monitor live diagnostic metrics, and handle maintenance outages safely.

- **Story 4.1: Manual Socket Status Override**
    - *Description*: As a Station Operator, I want to manually toggle a socket's status to "Out of Service" or "Maintenance", so that drivers cannot book malfunctioning hardware.
- **Story 4.2: Automated Maintenance Cancellation Notification**
    - *Description*: As a Station Operator, I want the system to automatically notify drivers and cancel active reservations if a charger goes offline unexpectedly, so that affected users can rebook at a nearby location.
- **Story 4.3: View Station Diagnostic Metrics**
    - *Description*: As a Station Operator, I want to view live hardware usage and diagnostic metrics for each socket, so that I can proactively schedule repairs before complete failure.

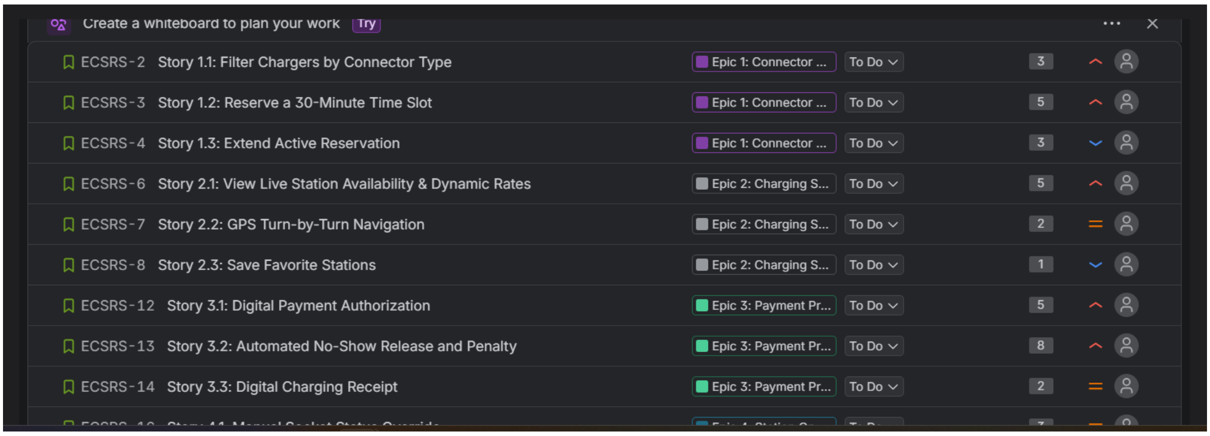
4. Visual Evidence & Jira Screenshots

Backlog

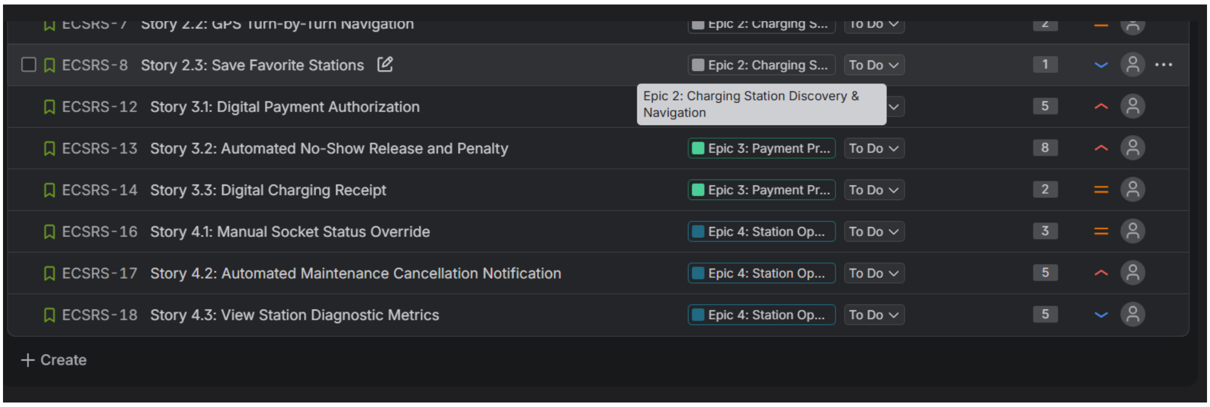


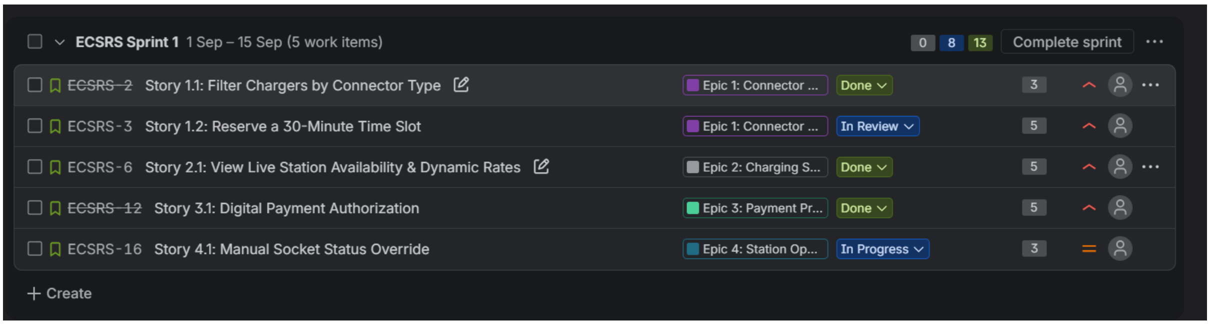


Story point assignment

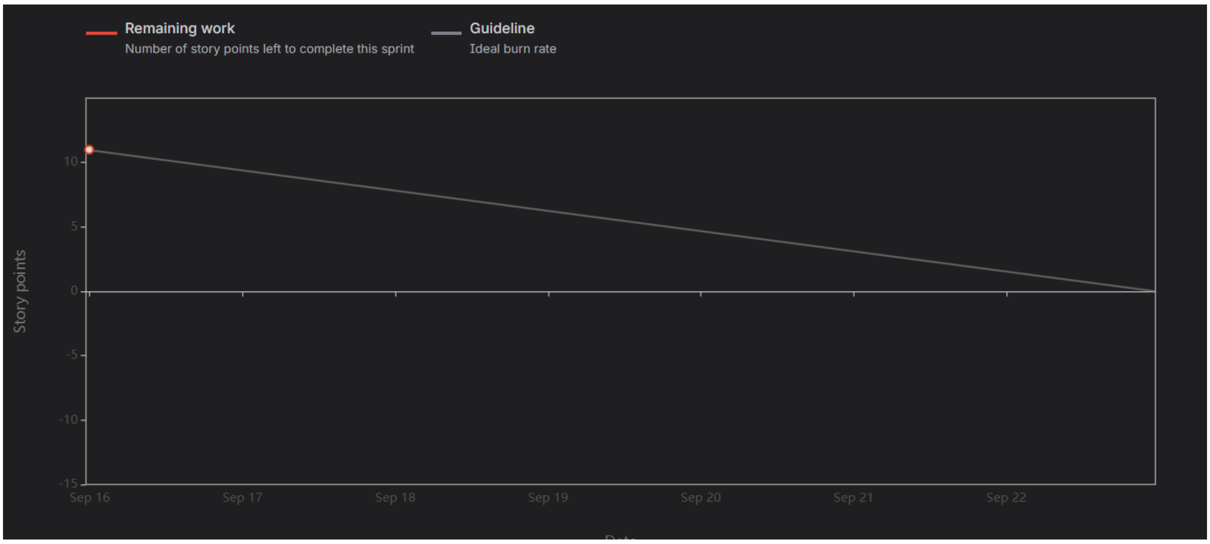


Active sprints





Burndown charts



5. Reflection & Sprint Analysis

1. Did your estimations reflect the actual effort?

Yes. Lower story point values (1–3 points) were assigned to self-contained tasks such as third-party navigation integration (Story 2.2) or local storage operations (Story 2.3). Higher story point values (5–8 points) accurately captured tasks with higher technical risks, such as database concurrency control for slot bookings (Story 1.2) and hardware sensor/OCPP state listeners for auto-releasing unfulfilled slots (Story 3.2).

2. Was your backlog well-prioritized?

Yes. Requirements essential for core product operations—such as filtering chargers, reserving slots, viewing live availability, and processing payments—were prioritized in Sprint 1. Auxiliary capabilities, such as favorite lists, diagnostics, and automated penalty releases, were prioritized in Sprint 2.

3. How did your simulated sprint align with your plan?

The planned sprint velocity of 21 story points for Sprint 1 provided a manageable scope. Tasks moved predictably through *To Do*, *In Progress*, and *Done*, ensuring that all core deliverables were finalized without bottlenecking.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart demonstrated a steady downward trajectory matching the ideal guideline curve. Step-down drops in remaining story points reflected task completions, indicating that the stories were well-scoped and broken down into manageable units.